BIO 304 . HUMAN ANATOMY & PHYSIOLOGY . WEEK 12 . MODULE 9 OF 17

Module 9: Special Senses

Topics: Vision, Hearing and Equilibrium

TOPIC 1 OF 2 . WEEK 12 . 6 CARDS (3 DOK 1 / 2 DOK 2 / 1 DOK 3)

Vision

Eye anatomy, accommodation, and photoreceptor function.

The Science

Light hits the cornea first; the cornea does most of the eye's fixed focusing. Then the lens, which is adjustable (accommodation). Then the vitreous humor; then the retina, which contains photoreceptors.

Accommodation: for near vision the ciliary muscle contracts, the zonular fibers slacken, and the lens elastically rounds up (stronger refraction). For far vision the ciliary muscle relaxes, zonules pull the lens flatter. With age the lens stiffens (presbyopia) and near vision worsens; reading glasses compensate.

Photoreceptors: rods are highly sensitive, give black-and-white vision, dominate at low light. Cones come in three types (red, green, blue), are less sensitive but give color and acuity, dominate in daylight. The fovea is the area of sharpest vision: all cones, no overlying neurons. The optic disc is where the optic nerve leaves the eye; no photoreceptors, hence the blind spot.

Phototransduction: light isomerizes the chromophore in rhodopsin (or cone opsins), which triggers a signaling cascade that hyperpolarizes the photoreceptor. Photoreceptors release less glutamate in light. Downstream bipolar and ganglion cells interpret the change. The signal exits via the optic nerve, partially crosses at the optic chiasm, and reaches the lateral geniculate nucleus and visual cortex.

Teaching This Topic

Before the video (drawing prompt)

Have students draw an eye in cross-section and label cornea, lens, retina, fovea, optic disc, optic nerve.

Common misconceptions to address

- The lens does most of the focusing. False: the cornea does most of it; the lens fine-tunes via accommodation.
- Rods give color vision. False: rods are achromatic; cones give color.
- Students think the blind spot is at the center of vision. It is offset; you do not notice it because the brain fills in the missing region.

Order of operations (what to teach first)

Light path through the eye, then accommodation, then photoreceptor types and distribution, then the basics of phototransduction.

Self-test prompts

- What structure does most of the eye's fixed refraction?
- Why is night vision mostly black and white?
- Why is there a blind spot and where is it?

Why This Matters to Your Patients

Cataracts cloud the lens (often from oxidative stress with age); surgery replaces the lens with a synthetic one. Glaucoma raises intraocular pressure (failure of aqueous humor drainage) and damages the optic nerve; nurses screen for this in older patients. Macular degeneration destroys the fovea, robbing central vision but sparing peripheral. Diabetic retinopathy damages retinal vessels and is a leading cause of blindness. Color blindness is usually an X-linked recessive disorder of cone opsins (mostly affecting males). Visual symptoms in any patient should prompt thinking about which anatomical level is failing.

TOPIC 2 OF 2 . WEEK 12 . 5 CARDS (3 DOK 1 / 1 DOK 2 / 1 DOK 3)

Hearing and Equilibrium

Cochlear transduction of sound and vestibular detection of head motion.

The Science

Sound path: pinna and external auditory canal funnel sound to the tympanic membrane. Vibration of the tympanic membrane moves the malleus, then the incus, then the stapes. The stapes pushes on the oval window, sending pressure waves into the perilymph of the cochlea.

In the cochlea, the basilar membrane vibrates, with different frequencies producing maximum vibration at different positions (tonotopy: high frequencies at the base, low frequencies at the apex). Hair cells in the organ of Corti have stereocilia that bend with basilar membrane motion. Bending opens mechanically gated K channels; the hair cell depolarizes; it releases glutamate onto CN VIII (the vestibulocochlear nerve), which carries the signal to the auditory cortex.

Equilibrium: the vestibule (utricle and saccule) detects linear acceleration and head tilt via otolith organs. The three semicircular canals detect rotational acceleration. All use hair cells with stereocilia in gel; head motion shears the gel and bends the cilia. Vestibular signals travel with auditory signals on CN VIII to the brainstem and cerebellum.

Teaching This Topic

Before the video (drawing prompt)

Have students draw the ear in cross-section: outer, middle, inner; tympanic membrane, ossicles, cochlea, semicircular canals. Then label them.

Common misconceptions to address

- Students think hearing happens in the cochlea entirely. It begins at the tympanic membrane and ends in the auditory cortex.
- All hearing loss is the same. Conductive loss (middle ear) is different from sensorineural loss (cochlea or nerve).
- Vertigo from inner ear problems is sometimes confused with light-headedness from low blood pressure. Vertigo is true spinning; light-headedness is not.

Order of operations (what to teach first)

Sound path mechanically, then cochlear transduction (tonotopy), then equilibrium as a parallel hair-cell system.

Self-test prompts

- Name the three middle ear ossicles in order.
- Where in the cochlea are high frequencies detected?
- Which structures detect rotational versus linear acceleration?

Why This Matters to Your Patients

Otitis media (middle ear infection) is the most common reason a child sees a doctor; fluid behind the tympanic membrane produces conductive hearing loss. Presbycusis (age-related sensorineural hearing loss) starts with high frequencies (hair cells at the cochlear base) and progresses. Benign paroxysmal positional vertigo (BPPV) is loose otoconia drifting into a semicircular canal; head movements trigger spinning vertigo. Meniere disease is endolymph dysregulation in the inner ear, producing episodic vertigo, fluctuating hearing loss, and tinnitus. Aminoglycoside antibiotics (gentamicin) are ototoxic; nurses monitor hearing carefully during long courses.